



Northeastern University
Khoury College of Computer Sciences
Computer Science Program
CS 5200 Database Management Systems

Final Exam (Mock)

Monday, April 20th, 2026

Name _____

ID _____

Duration: 90 minutes.

Question 1 10	
Question 2 10	
Question 3 10	
Question 4 10	
Question 5 10	
Question 6 10	
Question 7 20	
Question 8 20	
Total 100	

Instructions:

- ✓ Check to ensure that you have a complete copy of the examination. There are **8** Questions. There should be no blank pages in your set.
- ✓ This examination is open book and open notes.
- ✓ Questions are NOT allowed during the exam.
- ✓ Make sure to solve all Questions so that you get partial credit even if you do not finish a question.
- ✓ Include all steps and calculations as appropriate for maximum credit. Be clear, brief, and specific in your answers.
- ✓ If you find a mistake correct it
- ✓ If anything is missing add it

Good Luck!



Question #1

Specify the following queries on the COMPANY relational database schema shown in Figure 1, using the relational operators discussed in class.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	------------	-------	---------	-----	--------	-----------	-----

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
-------	----------------	---------	----------------

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
----------------	------------------

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
-------	----------------	-----------	------

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
-------------	------------	-------

DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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Figure 1: COMPANY relational database schema

1. Retrieve the names of employees in department 5 who work more than 10 hours per week on the 'ProductX' project.
2. Find the names of employees that are directly supervised by 'Franklin Wong'.
3. Find the names and addresses of employees who work on at least one project located in Houston but whose department has no location in Houston.



Answer to Question #1

1. EMP_W_X <-- (s PNAME='ProductX' (PROJECT)) J (PNUMBER),(PNO) (WORKS_ON)
EMP_WORK_10 <-- (EMPLOYEE) J (SSN),(ESSN) (s HOURS>10 (EMP_W_X))
RESULT <-- P LNAME,FNAME (s DNO=5 (EMP_WORK_10))
2. WONG_SSN <-- P SSN (s FNAME='Franklin' AND LNAME='Wong' (EMPLOYEE))
WONG_EMPS <-- (EMPLOYEE) J (SUPERSSN),(SSN) (WONG_SSN)
RESULT <-- P LNAME,FNAME (WONG_EMPS)
3. E_P_HOU(SSN) <-- P ESSN (WORKS_ON J(PNO),(PNUMBER) (s PLOCATION='Houston' (PROJECT)))
D_NO_HOU <-- P DNUMBER (DEPARTMENT) –
P DNUMBER (s DLOCATION='Houston' (DEPARTMENT))
E_D_NO_HOU <-- P SSN (EMPLOYEE J(PNO),(DNUMBER) (D_NO_HOU))
RESULT_EMPS <-- E_P_HOU - E_D_NO_HOU (* this is set DIFFERENCE *)
RESULT <-- P LNAME,FNAME,ADDRESS (EMPLOYEE * RESULT_EMPS)



Question #2

Show the results of each of the following queries when they are applied to the database whose state is in Figure 2

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Figure 2: State of COMPANY database

1. Retrieve the names of employees who work on every project.
2. For each department, retrieve the department name, and the average salary of employees working in that department.



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Answer to Question #2

1. Result (empty):
LNAME FNAME
2. Research 33250
Administration 31000
Headquarters 55000



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Question #3

Given the stored procedure shown in Figure 3, explain what it does assuming that it accesses the database shown in Figure 4 whose schema is shown in Figure 5.

```
loop=1;
while(loop)
{
    name = readEntry("Please enter your name: ");
    #sql iterator REPORT(char grade, double credit);
    REPORT r = null;
    #sql r =
    {
        select  G.Grade, C.Credit_hours
        from    STUDENT S, GRADE_REPORT G, SECTION SEC, COURSE C
        where   G.Student_number=S.Student_number AND
                G.Section_identifier=SEC.Section_identifier AND
                SEC.Course_number=C.Course_number AND S.Name=:name
    };
    count=0;
    sum=0;
    avg=0;
    while(r.next())
    {
        switch (r.grade)
        {
            case A: sum=sum+(4*r.credit); count=count+1; break;
            case B: sum=sum+(3*r.credit); count=count+1; break;
            case C: sum=sum+(2*r.credit); count=count+1; break;
            case D: sum=sum+(1*r.credit); count=count+1; break;
            case F: sum=sum+(0*r.credit); count=count+1; break;
            default: println("This grade "+r.grade+" will not be
calculated."); break;
        }

    };
    avg = sum/count;
    System.out.println("Student named "+name+" has a grade point average
"+avg+".");
    r.close();
}
```

Figure 3: Stored Procedure for Question 3



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STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 4: Database for Question 3

STUDENT

Name	Student_number	Class	Major
------	----------------	-------	-------

COURSE

Course_name	Course_number	Credit_hours	Department
-------------	---------------	--------------	------------

PREREQUISITE

Course_number	Prerequisite_number
---------------	---------------------

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
--------------------	---------------	----------	------	------------

GRADE_REPORT

Student_number	Section_identifier	Grade
----------------	--------------------	-------

Figure 5: Schema of Database for Question 3



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Answer to Question #3

program segment that reads a student's name and prints his or her grade point average, assuming that A = 4, B = 3, C = 2, and D = 1 points.



Question #4

Consider the library relational database schema in Figure 6. Write a program segment that retrieves the list of books that became overdue yesterday and that prints the book title and borrower name for each.

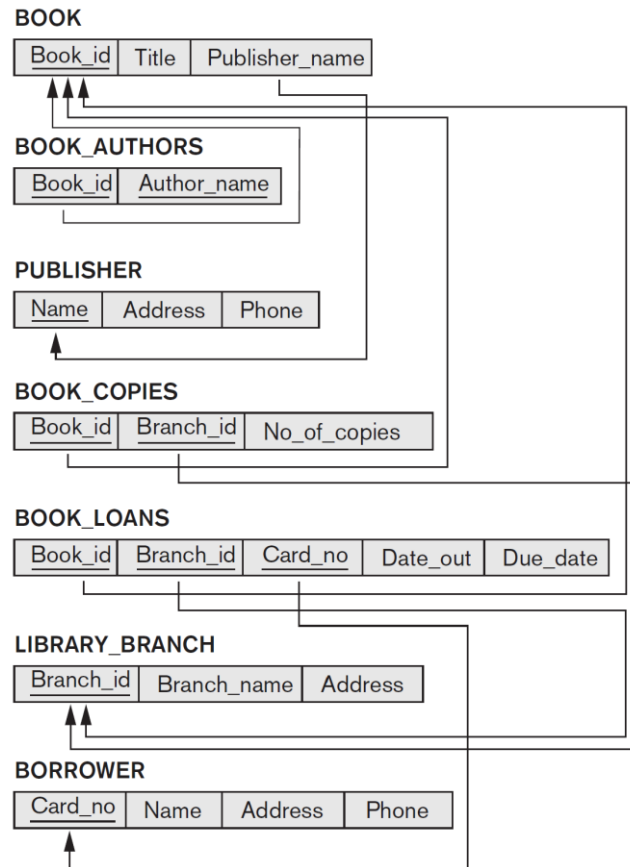


Figure 6: Relational Database Schema for Question 4

Answer to Question #4

```

#sql iterator BOOKLIST(String title, String borrowername);
BOOKLIST bl = null;
#sql bl =
{
    select  B.Title, BR.Name
    from    BOOK B, BOOK_LOANS BL, BORROWER BR
    where   B.Book_id=BL.Book_id AND BL.Card_no=BR.Card_no AND
           BL.Due_date={fn CURDATE() }-1
};
while(bl.next())
{
    System.out.println("Book titled "+bl.title+" was loaned by
"+bl.borrowername+".");
};
bl.close();
    
```



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Question #5

Indicate which schedule, in Figure 7, is serializable. If it is serializable, determine the equivalent serial schedule. If it is not serializable explain why it is not serializable and suggest a modification that will make it serializable.

$S1: r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$ $S2: r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$
--

Figure 7 Schedule for Question 5

Answer to Question #5

S1: This schedule is not serializable because T1 reads X ($r_1(X)$) before T3 but T3 reads X ($r_3(X)$) before T1 writes X ($w_1(X)$), where X is a common data item. The operation $r_2(X)$ of T2 does not affect the schedule at all so its position in the schedule is irrelevant. In a serial schedule T1, T2, and T3, the operation $w_1(X)$ comes before $r_3(X)$, which does not happen in the question.

S2: This schedule is serializable because all conflicting operations of T3 happens before all conflicting operation of T1. T2 has only one operation, which is a read on X ($r_2(X)$), which does not conflict with any other operation. Thus this serializable schedule is equivalent to $r_2(X); r_3(X); w_3(X); r_1(X); w_1(X)$ (e.g., T2 ,T3 ,T1) serial schedule.



Question #6

Draw the serializability graph and indicate which schedule, in Figure 8, is serializable. If it is serializable, determine the equivalent serial schedule. If it is not serializable explain why it is not serializable and suggest a modification that will make it serializable.

$T1: r_1(x); r_1(z); w_1(x)$ $T2: r_2(z); r_2(y); w_2(z); w_2(y)$ $T3: r_3(x); r_3(y); w_3(y)$ $S1: r_1(x); r_2(z); r_1(x); r_3(x); r_3(y); w_1(x); w_3(y); r_2(y); w_2(z); w_2(y)$ $S2: r_1(x); r_2(z); r_3(x); r_1(z); r_2(y); r_3(y); w_1(x); w_2(z); w_3(y); w_2(y)$
--

Figure 8 Schedule for Question 6

Answer to Question #6

Schedule S1: It is a serializable schedule because

- T1 only reads X ($r_1(X)$), which is not modified either by T2 or T3,
- T3 reads X ($r_3(X)$) before T1 modifies it ($w_1(X)$),
- T2 reads Y ($r_2(Y)$) and writes it ($w_2(Y)$) only after T3 has written to it ($w_3(Y)$)
- Thus, the serializability graph is



Schedule S2: is not a serializable schedule because

- T2 reads Y ($r_2(Y)$), which is then read and modified by T3 ($w_3(Y)$)
- T3 reads Y ($r_3(Y)$), which then modified before T2 modifies Y ($w_2(Y)$)

In the above order T3 interferes in the execution of T2, which makes the schedule nonserializable.

